Varsha Shukla

DOCTOR OF PHILOSOPHY STUDENT AT DARTMOUTH COLLEGE

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I am a Doctor of Philosophy Student at the Thayer School of Engineering, Dartmouth College. Research interests include Generative AI, Large Language Models, Machine Learning, Artificial Intelligence, Exploratory Data Analysis, and Platform Economics & Strategy. Prior to this, I was working as the Head of Engineering with the Ministry of Railways, Government of India. My experiences here have given me a strong foundation in Operations, Automation, Digital Customer Access, Big Data Analytics, Public Policy, Platform Network Effects, and Technology Development.

Work Experience

Summer Internship at Sight Machine, San Francisco, California, United States **2023-Present**

- Collaborated directly with the CTO of Sight Machine, Dr. Kurt DeMaagd to delve into cutting-edge technologies, including Generative AI for developing personalized large language model prototypes.
- Spearheaded exploratory data analysis initiatives, extracting actionable insights to enhance decision-making for the fast-growing startup.
- Contributed valuably to refining AI algorithms and strategies, showcasing a proactive approach to innovation and teamwork.

Head of Engineering, Indian Engineering Services, Ministry of Railways, Government of India, India **2017-2022**

- Appointed by the President of India in the Premium Indian Engineering Services. Secured All India Rank 4th in the examination conducted by Union Public Service Commission, Government of India among 30,00,000+ college graduates in the Prestigious National Engineering Services Examination.
- Directed Technology Development projects worth $300 million, completing them in a quarter of the estimated time. My work was recognized by the Chairman Railway Board for efficiency and timely execution. I have worked with teams in Russia, Japan, Vienna, and Germany to transform Indian Railways Digitally.
- Implemented Predictive Maintenance as a Service (PMaas) for Rolling Stock Predictive Maintenance by consolidating volumes of maintenance data with analytics using cloud computing and improving remote diagnosis and fault analysis using automatic data visualization.
- Worked on mobility as a service (MAAS) solutions like making applications to analyze different journey scenarios, ticketing service, and monitoring rail traffic to prevent rail accidents, road works, incidents and accidents online. Brought Grades in Automation in Delhi Metro such as driverless metro lines, light rail transit (LRT), people mover, and automated guided transit.

Entrepreneurial Lead, NSF iCorps, Cornell University, and University of Pennsylvania **2023-Present**

- NSF-led I-Corps program offers immersive, entrepreneurial training for students, faculty, post-docs, and other researchers interested in exploring the market potential of their work.
- Through the IN I-Corps Hub, Dartmouth I worked to explore market validation for my startup and had access to intellectual and entrepreneurial tools and resources.

Platform Strategy Consultant, Lockheed Martin Corporation, United States **2022-2023**

- Developed a scalable architectural framework and Platform Design Model (MVP) of the Cloud-Based data repository for Tech Strategy relating to small businesses in the additive manufacturing and digital twin domain.
- Determined and designed different user interfaces depending on the identity of access being internal or external and carried out a return-on-investment analysis for this effort in labor hours.

Vice President, Client Outreach and Network, Dartmouth Graduate Consulting Group **2022-Present**

- Worked with clients in the Upper Valley and nearby regions to develop their growth and marketing strategy.
- Conduct Initial Interviews with clients to understand client's needs and expected deliverables and timeline.
- Cultivate a strong relationship with clients through monthly networking events and connections throughout.

Graduate Teaching Assistant, Thayer School of Engineering, Dartmouth College **2022-Present**

- Graduate Teaching Assistant to the Master's Grad at Thayer and Tuck School for Platform Design and Management.
- Subjects studying Platform Design, Management & Strategy, Game Theoretic Design, Learning and Engineering, Operations Research, Bayesian Statistical Modelling, and Computation and Decision-Making under Uncertainty.
- Innovation Program Courses such as Technology Innovation and Entrepreneurship, Accounting and Finance, Law for Technology and Entrepreneurship, and Introduction to Innovation.

Education History

Dartmouth College-Thayer School of Engineering - Doctor of Philosophy in Engineering Sciences Ongoing

- Pursuing Ph.D. at Thayer School of Engineering, Dartmouth College in Engineering Sciences
- Research focussed on Generative AI, Large Language Models, and Platform economics & Strategy.

Indian Institute of Technology, Kanpur - Dual, Electrical Engineering & Power Engineering 2017

- Department of Electrical Engineering & Power Engineering Graduated Honors 8.31 CGPA
- Awarded Professor L. P. Singh Power System Research Award for Best Research Work

Invited Talks

https://youtu.be/XEisjCWOVw0?t=982 Experience Sharing with the President of India: Talk by top officers of the Ministry of Railways called upon by the President of India, Shri Ram Nath Kovind in Durbar Hall, Rashtrapati Bhawan

Academic Awards

All India Rank 4th Indian Engineering Services Awarded All India Rank 4 in Indian Engineering Services Conducted by Union Public Service Commission
Professor L.P Singh Best Research Thesis Award Professor L. P. Singh Power System Research Award for Best Research Thesis Work carried out in Electrical Engineering
Distinguished Alumnus Award For the Year 2018 Awarded by the University to acknowledge the success of students who have distinguished themselves in their field.
Academics Excellence Awards For All Years Best in Academics Award in every Semester in Bachelors Academic Excellence is awarded to the Topper of the Batch.

Publication

On adaptive fault current limiter in systems with distributed generation Published an IEEE paper on adaptive protection devices solving the problem of RES DG integration into the distribution system https://ieeexplore.ieee.org/document/8070005

Patent

U2024-007 / Patent Assignment / U.S. Application No. 63/543,282 / Advanced AI-Driven Personalized Large Language Model for Innovative Commercial Solutions/ Shukla / 094219.00039

Foreign Experiences

St Petersburg State Transport University
Specialized in railway transport. It is the first higher transport and construction educational institute in Russia. Learned about the mechanics and communication in railway transport, electricity railways, Ecology on the railways and in the urban economy, transport construction, high-speed movement, and Transportation cities.

Moscow State University of Railway Engineering
Went for a term as a part of our training with the Ministry of Railways. Studied transportation systems, technology and organization of production, cargo, and commercial work, saving technologies in rail transport, studies of bridges and transportation facilities, economic problems in railway transport, the safety of the transportation process, information of transport, track and track facilities, railway rolling stock and automatics.

Mexico City Social Entrepreneur Experience
It provided an opportunity to immerse oneself in the world of social entrepreneurship and sustainability. Eco-building initiative at Ehecalli got us hands-on and contribute to the community and taught us a collaborative and community-driven approach to problem-solving. Visits to places like the Nacional de Antropología and New Comienzos taught us different cultures and diverse perspectives and also taught us to be adaptable and open-minded, which are valuable traits in life and work.

Associations and Conferences

MIT Initiative on the Digital Economy: The MIT Initiative on the Digital Economy explored the transformative potential of digital innovation, examining its impact on the economy, business, and society at large.

Boston University Maching Market: BU Maching Market offered insights into entrepreneurship and the startup ecosystem, focusing on practical knowledge and strategies for launching and scaling new ventures.

Boston University Platform Symposium: The BU Platform Symposium delved into emerging technologies and their societal implications, highlighting the ethical considerations and responsible development of technological innovations.

Global Health Fellows: The Global Health Fellows focused on exploring global health challenges, innovative solutions, and the importance of cross-disciplinary collaboration in addressing public health issues worldwide.